



Department of Electronic & Telecommunication Engineering, University of Moratuwa, Sri Lanka.

Assignment 1 : Report

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BM4152 Bio Signal Processing

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1 Smoothing Filters

1.1 Moving Average Filter

The moving average (MA) filter of order N is characterized by the difference equation:

$$y(n) = \frac{1}{N} \sum_{k=0}^{N-1} x(n-k) \quad (1)$$

This filter is commonly used to remove high-frequency noise from biosignals. In this section, a clean ECG signal is loaded and intentionally corrupted with additive white Gaussian noise (AWGN) to analyze filter performance.

1.1.1 Preliminaries

The clean ECG signal is loaded from `ECG_template.npz` with a sampling frequency of $f_s = 500$ Hz and amplitude units in mV. Figure 1 depicts a single pulse of the clean ECG signal.

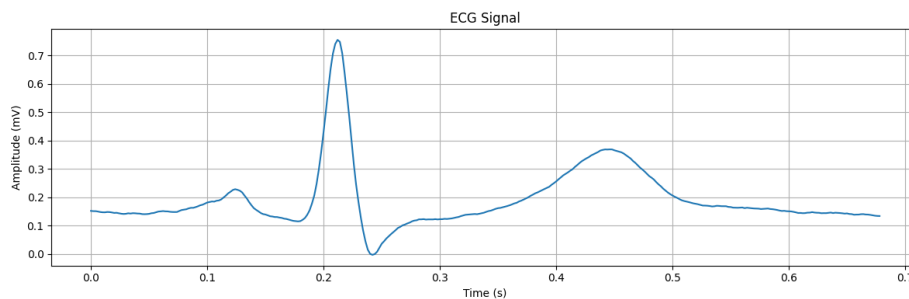


Figure 1: The waveform data depicting a single pulse of the ECG signal.

The clean signal is corrupted with AWGN to achieve a target SNR of 5 dB, producing the noisy signal denoted as `nECG`. Figure 2 shows the noisy ECG waveform, and its power spectral density (PSD) estimate is shown in Figure 3, confirming wide-band noise addition.

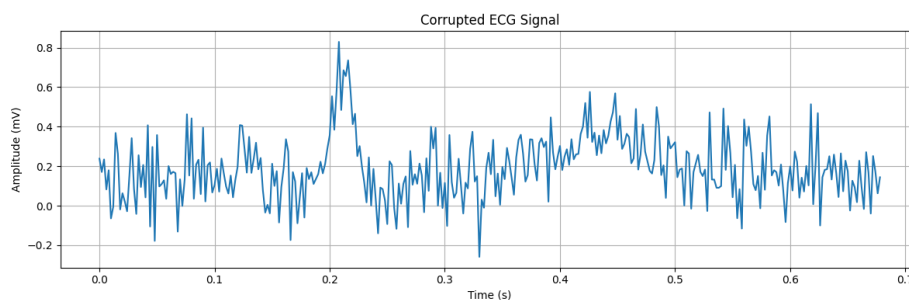


Figure 2: Noisy ECG signal after adding AWGN with 5 dB SNR.

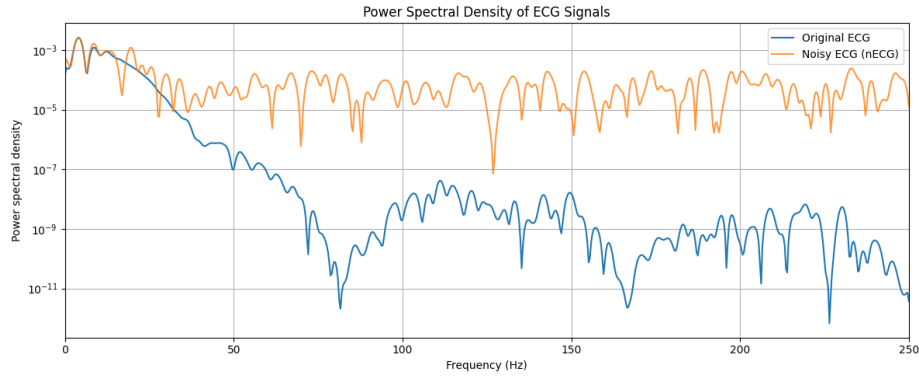


Figure 3: Power Spectral Density (PSD) estimate of the noisy ECG signal.

1.1.2 MA(3) Filter Implementation

A custom script implements an MA(3) filter ($N = 3$). The group delay for this symmetric FIR filter is calculated as:

$$T_g = \frac{N - 1}{2} \quad (2)$$

Accounting for a group delay of 1 sample, delay compensation is applied to the output. Figure 4 displays the filtered signal alongside the noisy signal, while Figure 5 compares their PSD estimates, highlighting the suppression of frequency components above 100 Hz.

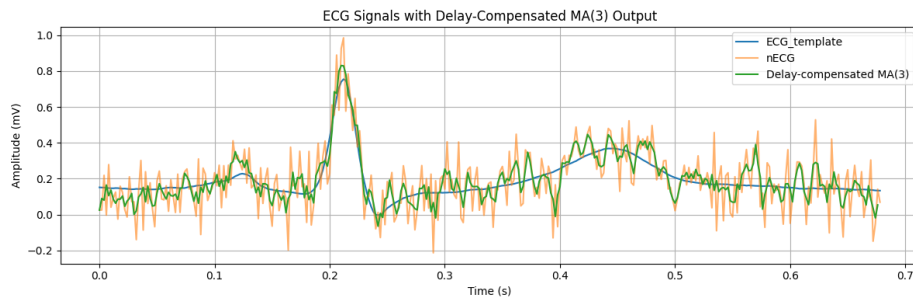


Figure 4: Signal filtered using a custom-written MA(3) filter.

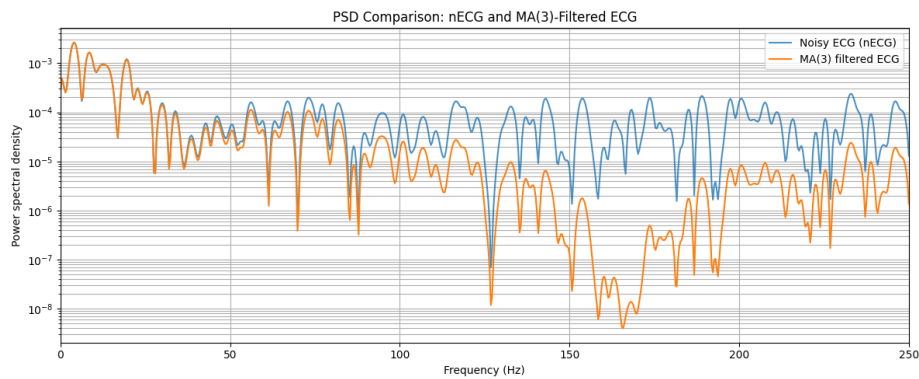


Figure 5: Power Spectral Density comparison of the signal filtered using custom MA(3).

Using the built-in filtering command with $N = 3$, the output is delay-compensated and visualized

in Figure 6. The frequency response, phase response, and pole-zero distribution are analyzed in Figures 7 and 8. The filter has two poles located at $z = 0$, while its zeros are located at $z = -0.5 + j0.8660254$ and $z = -0.5 - j0.8660254$.

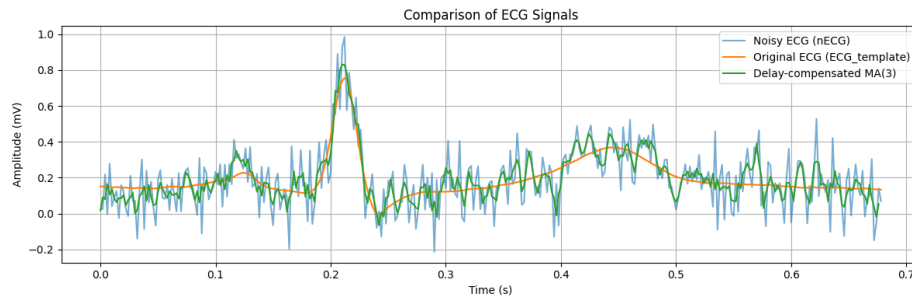


Figure 6: Signal filtered using the built-in MA(3) command.

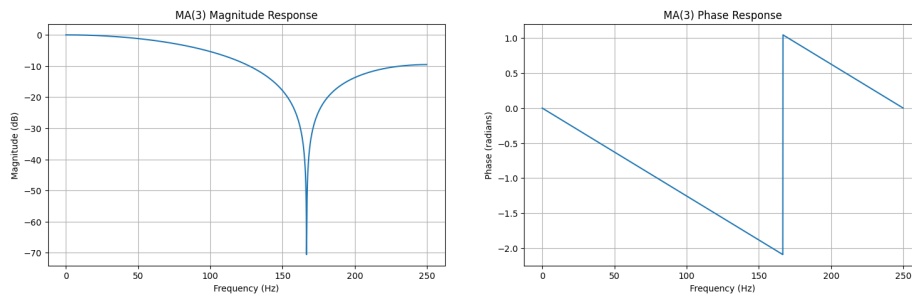


Figure 7: Magnitude and phase responses for the MA(3) filter.

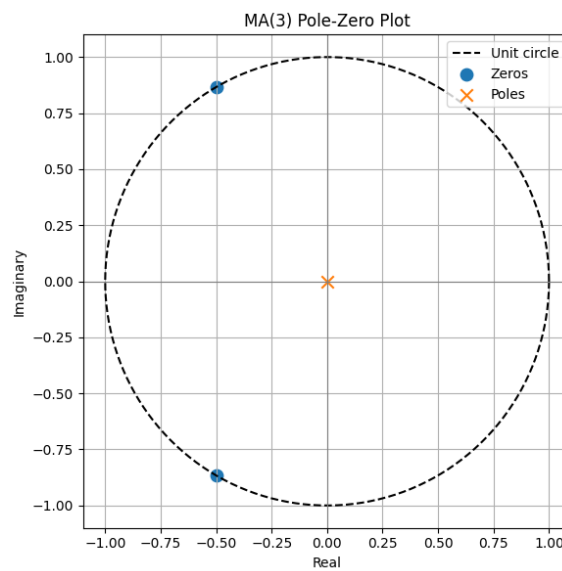


Figure 8: Pole-zero plot of the MA(3) filter.

1.1.3 MA(10) Filter and Optimum Order Selection

Increasing the filter order to $N = 10$ provides enhanced high-frequency attenuation, as observed in its magnitude response, phase response and pole-zero plot (Figure 9) and filtered signal output (Figure 12). The higher-order moving average filter provides improved smoothing of the ECG signal by averaging over a larger number of samples. This results in stronger suppression of high-frequency components, including the white noise present in the signal, producing a cleaner and smoother output compared with the lower-order filter.

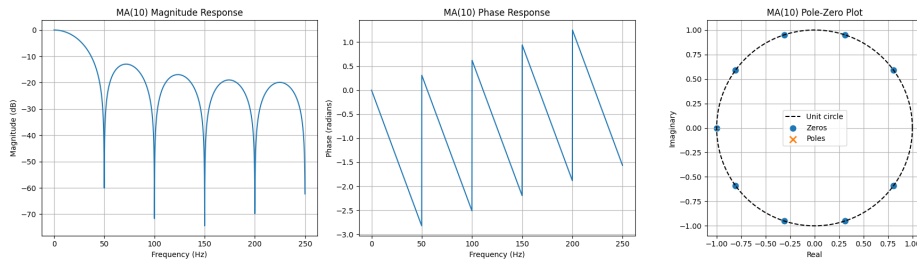


Figure 9: Magnitude, Phase response, and Pole-Zero of the MA(10) filter.

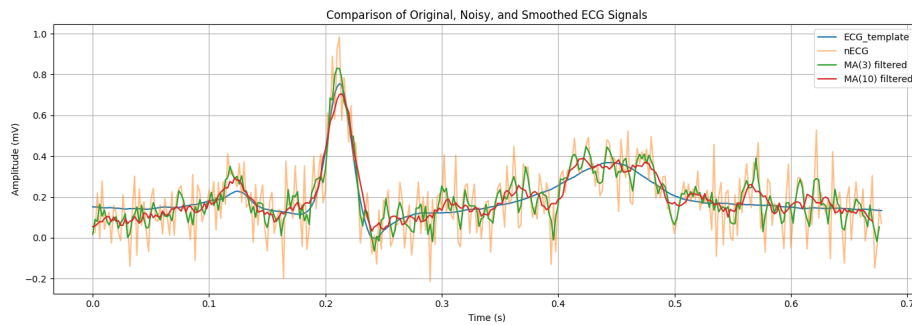


Figure 10: Filtered signal using the MA(10) filter.

To determine the optimal filter order N , the Mean Squared Error (MSE) between the filtered signal and the clean signal is computed across a range of N values (Figure 11). The minimum MSE is achieved at $N = 12$.

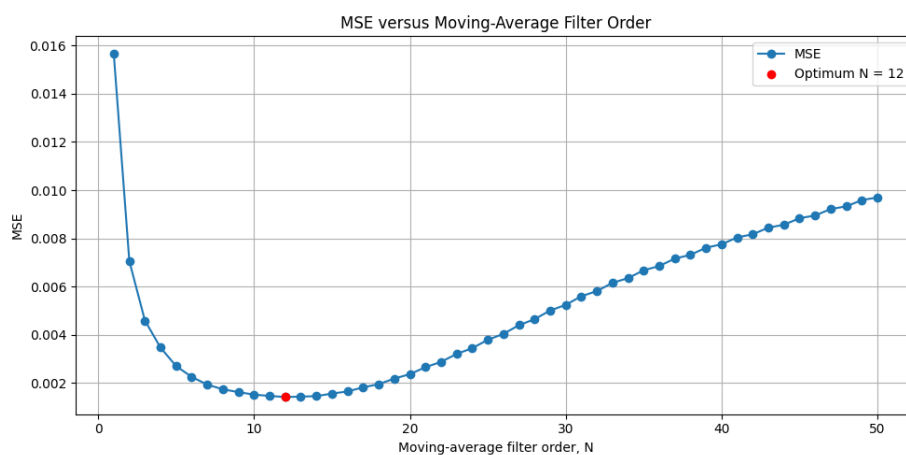


Figure 11: Mean Squared Error against filter order N , showing a minimum at $N = 13$.

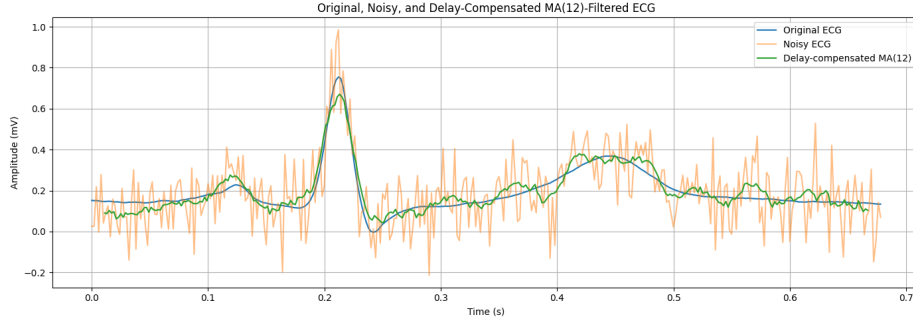


Figure 12: Filtered signal using the MA(12) filter.

Analysis of Filter Order Extremes: At very low filter orders, the MA filter fails to sufficiently smooth the signal, allowing substantial high-frequency noise to remain and resulting in a high MSE. Conversely, excessively high orders oversmooth the ECG waveform, distorting critical transient features like sharp QRS complexes and once again elevating the MSE. An intermediate order ($N = 12$) maintains the optimal trade-off between noise reduction and signal fidelity.

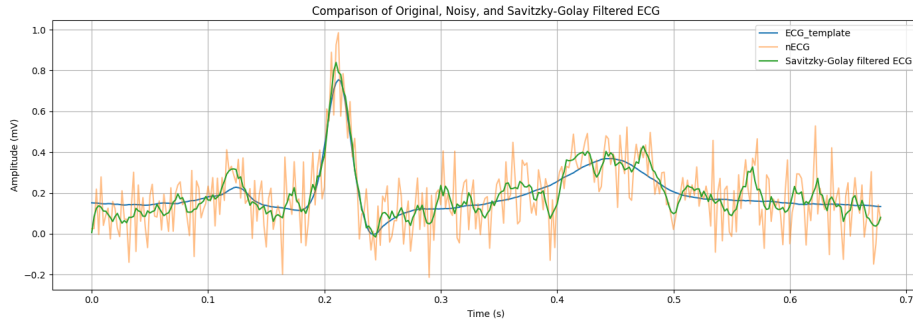
1.2 Savitzky-Golay Filter

The Savitzky-Golay (SG) filter fits a least-squares polynomial of order N over a sliding window of length $L' = 2L + 1$:

$$y(n) = \sum_{k=0}^N a_k n^k \quad (3)$$

where $N \leq L' - 1$ must be satisfied to ensure a unique solution.

Applying an $SG(3, 11)$ filter yields the output shown in Figure 13. While noise is effectively reduced compared to lower-order MA filters, flat regions tend to exhibit polynomial curvature rather than remaining flat.

Figure 13: Output of the Savitzky-Golay filter applied with $N = 3$ and $L' = 11$.

To locate the optimal parameters, the MSE is evaluated across variations of N and window half-width L (Figure 14). The minimum MSE corresponds to $N = 6$ and $L = 26$, producing the optimally filtered signal shown in Figure 15.

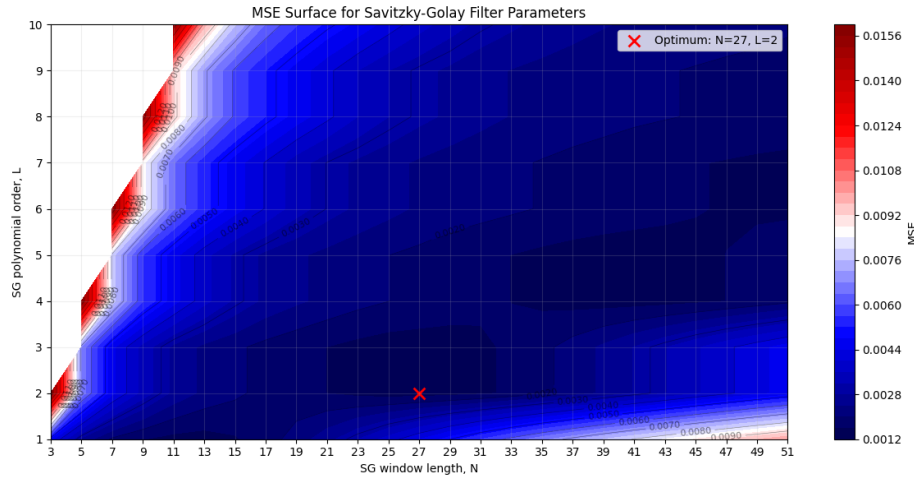


Figure 14: Mean Squared Error variation against polynomial order N and window half-width L .

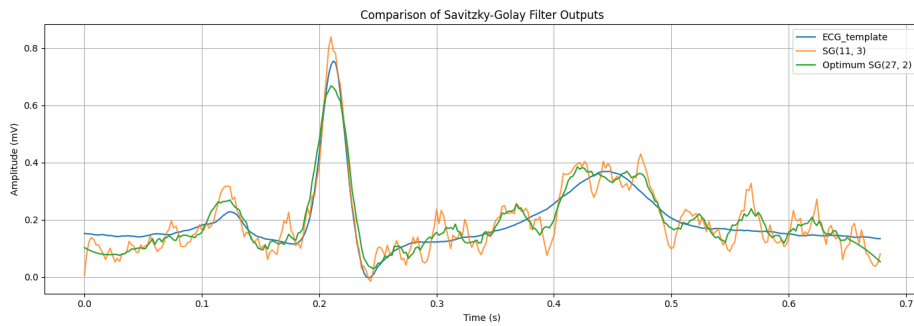


Figure 15: Optimally filtered signal using the Savitzky-Golay filter ($N = 6, L = 26$).

1.3 Comparison of Moving Average and Savitzky-Golay Filters

- **Computational Complexity:** MA filters are computationally efficient due to constant coefficients requiring only simple additions and divisions, making them ideal for real-time systems. SG filters require polynomial least-squares fitting per data window, making them more computationally demanding and better suited for offline analysis.
- **Signal Features:** MA filters tend to oversmooth signals and distort sharp QRS complexes at higher orders. In contrast, SG filters preserve local waveform morphology and sharp slopes more effectively while suppressing high-frequency noise.

2 Ensemble Averaging

In scenarios where the power spectra of the signal and noise overlap significantly, traditional filtering methods are often ineffective because they attenuate desired signal components alongside the noise. An alternative approach is synchronized ensemble averaging, which exploits the assumption that the desired signal is repeatable across trials while the noise is random and uncorrelated. Averaging across multiple measurements causes the noise to cancel out while reinforcing the time-locked signal components.

Mathematically, let the i -th recorded signal be expressed as:

$$y_i(n) = x(n) + \eta_i(n) \quad (4)$$

where $x(n)$ is the deterministic, trial-invariant desired signal and $\eta_i(n)$ represents the random noise component. The ensemble average over M trials is given by:

$$\bar{y}(n) = \frac{1}{M} \sum_{i=1}^M y_i(n) \quad (5)$$

Given that $\mathbb{E}[\eta_i(n)] = 0$, the average converges to the true underlying signal as $M \rightarrow \infty$:

$$\lim_{M \rightarrow \infty} \bar{y}(n) = x(n) \quad (6)$$

2.1 Signal with Multiple Measurements (ABR)

This experiment examines the Auditory Brainstem Response (ABR), an early component of auditory evoked potentials generated in response to auditory stimuli and recorded via scalp EEG electrodes. While ABR amplitudes are in the microvolt (μV) range, the signals are heavily contaminated by background EEG activity in the millivolt (mV) range. Ensemble averaging over numerous trials significantly improves the signal-to-noise ratio (SNR) to reveal the underlying brainstem response.

2.1.1 Preliminaries and Implementation

The dataset loaded from `ABR_rec.npz` comprises over 1000 ABR trials sampled at $f_s = 40 \text{ kHz}$, with epochs spanning 10 ms post-stimulation. The recording contains two columns: the first represents the auditory stimulus (a 0.1 ms pulse at $f = 21.1 \text{ Hz}$), and the second contains the recorded ABR mixed with EEG noise.

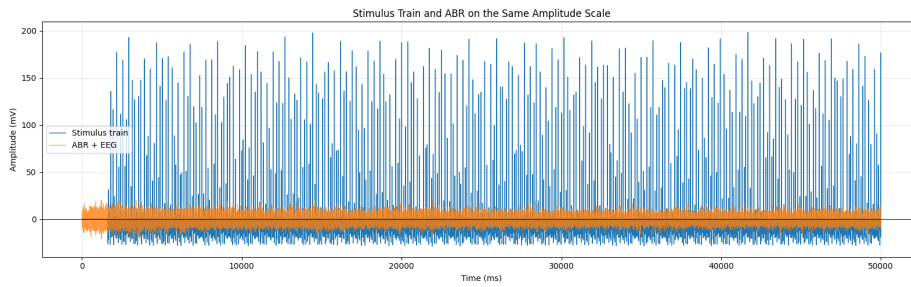


Figure 16: Pulse train stimuli and corresponding ABR recording data.

Stimulus occurrences were detected by applying a voltage threshold of 50 to the stimulus channel, yielding 1034 identified stimulus events after filtering out consecutive redundant detections. Using these reference time points, individual ABR epochs were segmented with a fixed 12 ms window (-2 ms to $+10 \text{ ms}$, corresponding to sample indices -80 to 399 at $f_s = 40 \text{ kHz}$). Computing the ensemble average across these epochs suppresses uncorrelated EEG noise while reinforcing the consistent ABR waveform, producing the clean time-domain trace shown in Figure 17.

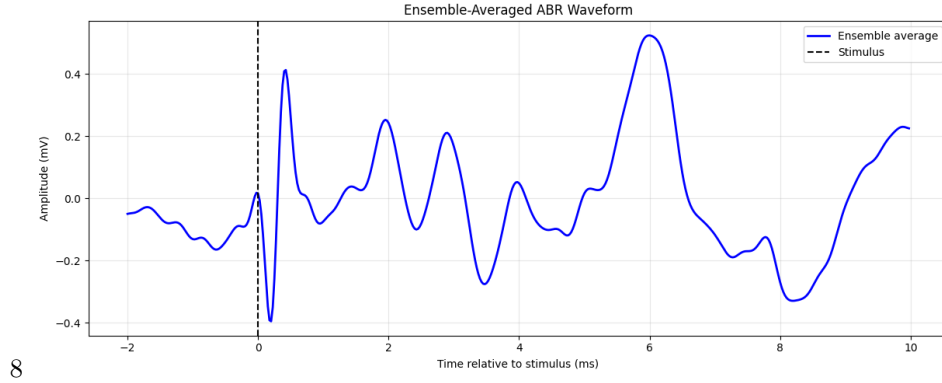


Figure 17: Ensemble-averaged ABR waveform derived from the recorded epochs.

2.1.2 Improvement of the SNR

Using the ensemble-averaged ABR as a fixed template, progressive averages were computed from an increasing number of single-trial epochs. The Mean Squared Error (MSE) between each progressive average and the template was evaluated for all $k = 1, 2, \dots, M$ according to the formulation:

$$\text{MSE}_k = \frac{1}{N} \sum_{n=1}^N (y(n) - \hat{y}_k(n))^2 \quad (7)$$

where $y(n)$ is the template, N is the length of a single ABR epoch, M is the total number of epochs, and $\hat{y}_k(n)$ represents the progressive average of k epochs defined as:

$$\hat{y}_k(n) = \frac{1}{k} \sum_{i=1}^k x_i(n) \quad (8)$$

Figure 18 illustrates the variation of MSE against the number of epochs k . As expected for time-locked responses contaminated with uncorrelated background noise, the curve demonstrates a monotonic decrease in MSE as k increases, directly reflecting the theoretical improvement of SNR with ensemble size.

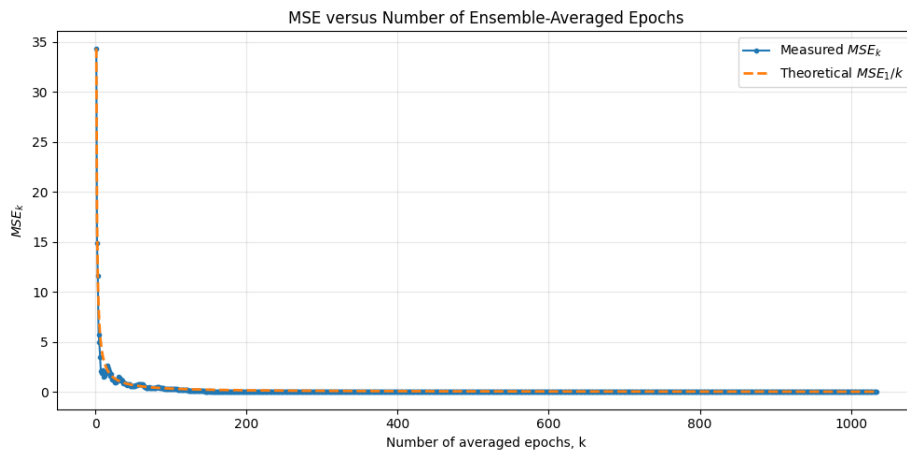


Figure 18: Mean Squared Error against the number of epochs k .

2.2 Signal with Repetitive Patterns

An almost periodic ECG pulse train recorded at a sampling frequency of $f_s = 128$ Hz (Lead II, with amplitude units in mV) is considered to study ensemble averaging. To emulate practical recording conditions, Gaussian white noise is added to the signal, causing an overlap between the ECG spectrum and the noise spectrum.

The acquisition parameters for this dataset (`ECG_rec.npz`) are summarized in Table 1.

Property	Value
Name of recording	<code>ECG_rec.npz</code>
Recorded Lead	Lead II
Sampling frequency	128 Hz
Amplitude units	mV

Table 1: Signal Properties for Repetitive Pattern ECG Recording.

The loaded ECG signal is visualized in the time domain to observe its baseline waveform and amplitude features (Figure 19). Before contaminating the signal with additive noise, a single PQRST waveform is extracted to serve as a reference template (`ECG_template`) for subsequent epoch segmentation via cross-correlation (Figure 20).

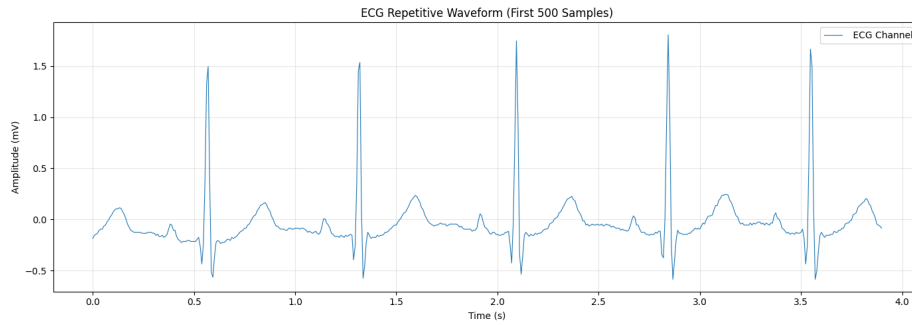


Figure 19: Loaded ECG pulse train recording.

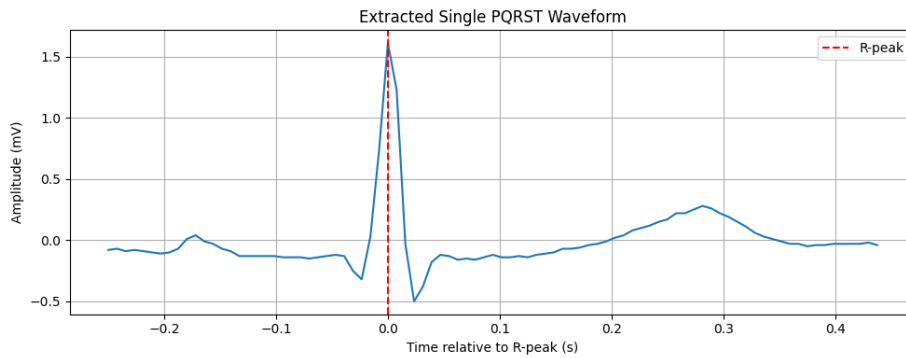


Figure 20: Extracted single PQRST waveform serving as `ECG_template`.

Additive white Gaussian noise is then introduced to the signal at an SNR of 5 dB, generating the noisy signal sequence designated as `nECG` (Figure 21).

To segment the noisy ECG pulse train into individual epochs, normalized cross-correlation is computed between the clean `ECG_template` and `nECG`. The resulting correlation values are plotted against a lag axis converted to time (Figure 22).

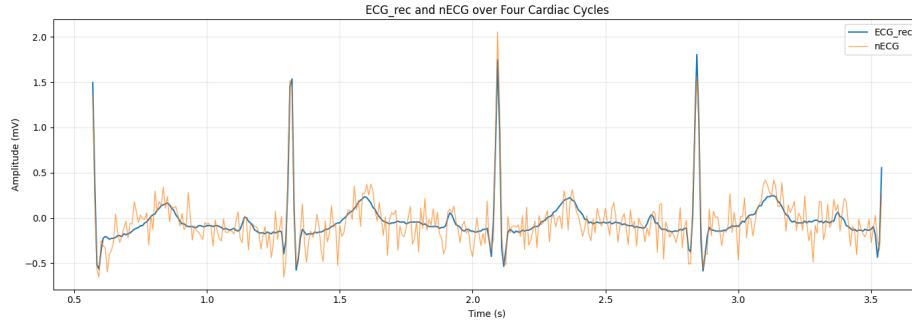


Figure 21: Complete ECG pulse sequence contaminated with 5 dB Gaussian white noise (nECG).

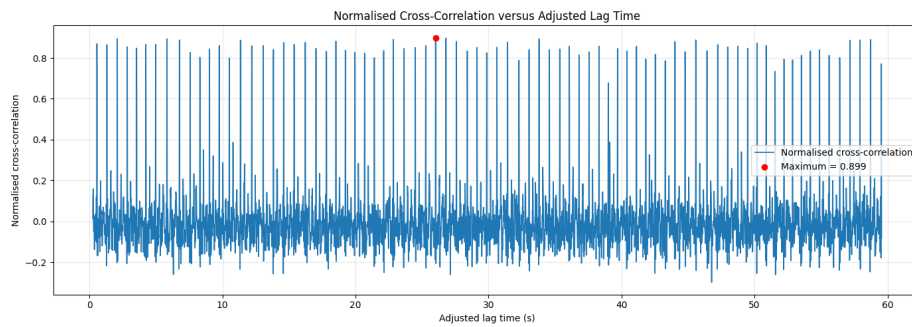


Figure 22: Normalized cross-correlation between the template and the noisy ECG sequence.

A threshold is applied (0.7 was used here) to the cross-correlation peaks to identify individual pulse onsets, allowing fixed-length epochs to be segmented and ensemble averaged (Figure 23).

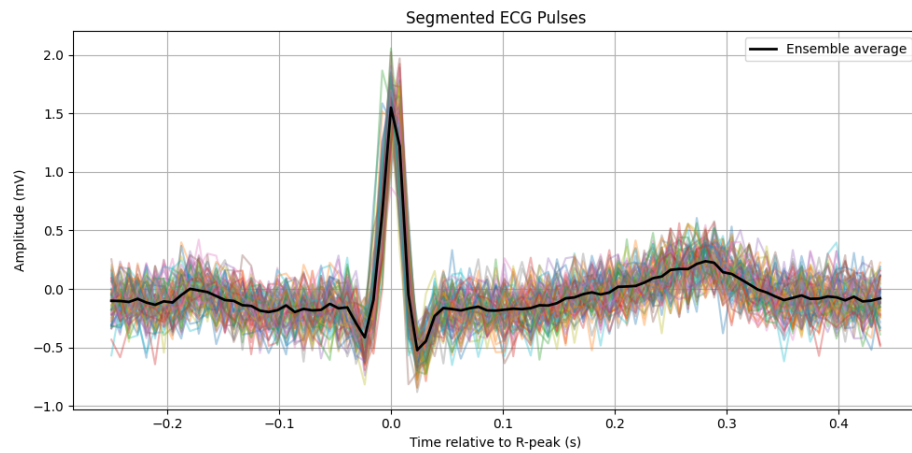


Figure 23: Sample segmented ECG epochs using cross-correlation peaks being ensemble averaged.

Ensemble averages are formed by cumulatively increasing the number of included pulses, and the resulting signal-to-noise ratio (SNR) is evaluated and plotted against the epoch number k (Figure 24).

To visualize the noise reduction efficacy, two ensemble averages corresponding to $k = 5$ and $k = 78$ epochs are plotted alongside a single noisy ECG pulse (Figure 25).

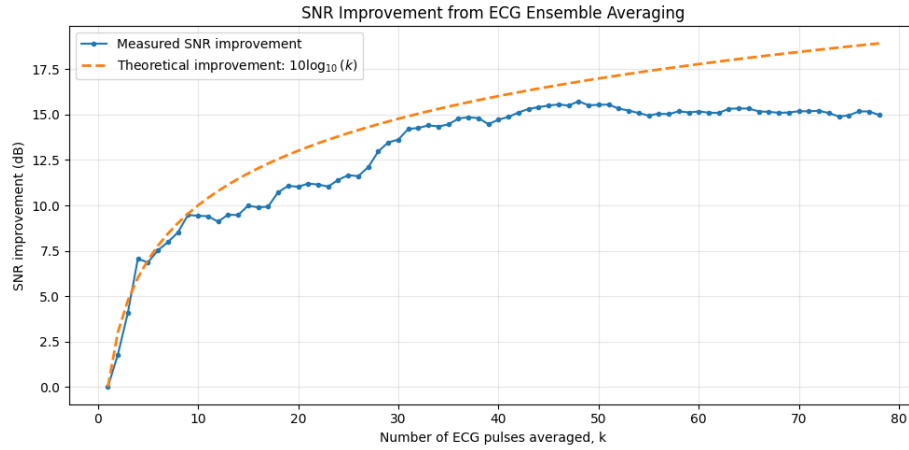


Figure 24: Signal-to-noise ratio (SNR) improvement against the number of epochs k .

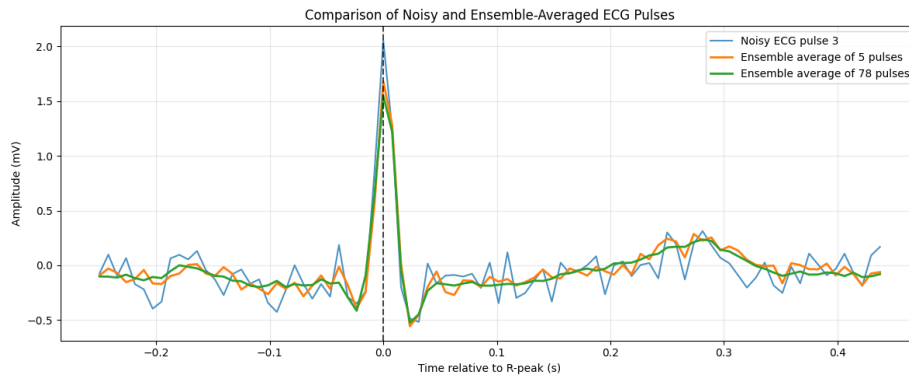


Figure 25: Comparison of a single noisy ECG pulse against ensemble-averaged pulses for $k = 5$ and $k = 78$.

Maximum Correlation Points vs. R-Peak Detection: Utilizing points of maximum cross-correlation with a template provides a more reliable segmentation strategy than isolating individual R-peaks. Because cross-correlation evaluates the entire PQRST morphology rather than a single local extremum, it significantly mitigates the risk of mis-detections in heavy noise environments where R-waves may be distorted or obscured. This ensures precise temporal alignment of each extracted heartbeat, yielding superior ensemble averaging performance and enhanced noise suppression.

3 Design of FIR Filters Using the Windowing Method

3.1 Characteristics of Window Functions

The rectangular window is the simplest window function. Varying its length (M) directly impacts the time-domain impulse response and the resulting frequency resolution. Figure 26 overlays the impulse responses for $M = 5, 50$, and 100 . As M increases, the window widens in the time domain, which in turn narrows the main lobe and improves frequency resolution in the spectral domain.

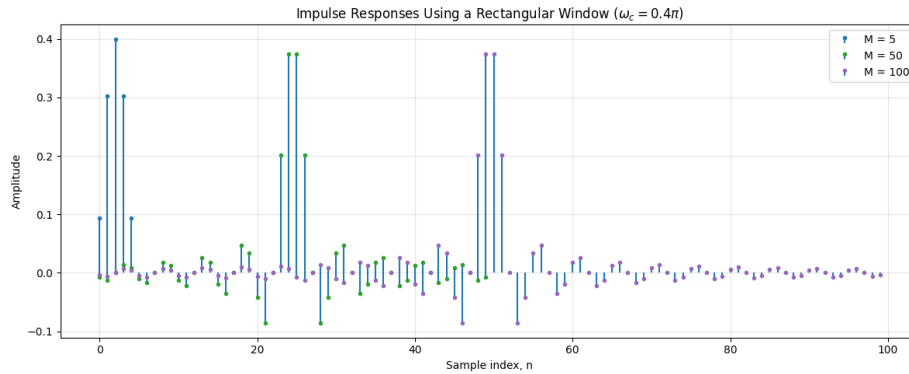


Figure 26: Impulse response of the rectangular window filter for $M = 5, 50, 100$.

3.1.1 Comparative Response for Lowpass Filter ($\omega_c = 0.4\pi$)

A lowpass filter with a normalized cutoff frequency of $\omega_c = 0.4\pi$ is designed across filter orders $M = 5, 50$, and 100 . Figures 27 illustrate the comparative linear magnitude responses and phase responses, respectively. Larger window lengths produce sharper transition bands and enhanced frequency selectivity, whereas lower orders exhibit highly smoothed, gradual roll-offs.

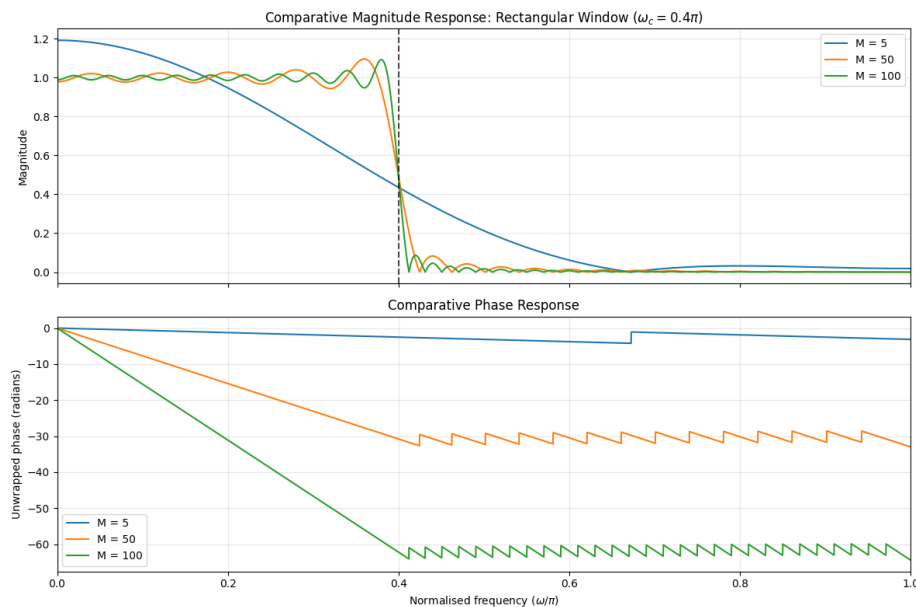


Figure 27: Magnitude and phase responses of the lowpass FIR filters for orders $M = 5, 50, 100$.

3.1.2 Comparison of Window Functions ($M = 50$)

To mitigate the high side-lobes caused by the abrupt truncation (discontinuities) of the rectangular window, smoother window functions are introduced. Rectangular, Hanning, Hamming, and Blackman windows of length $M = 50$ are compared across their morphological and spectral properties:

- **Morphology of the Window:** Smooth windows taper smoothly toward zero at their boundaries, reducing edge discontinuities compared to the abrupt rectangular profile.
- **Magnitude Response (Linear Scale):** The linear magnitude responses (Figure 29) demonstrate how tapered windows widen the main transition band while suppressing side-lobe ripples.
- **Magnitude Response (Logarithmic Scale):** Logarithmic plots (Figure 30) highlight the trade-off in stopband attenuation, where windows like Blackman provide deep attenuation at the expense of a wider transition region.
- **Phase Response:** All linear-phase symmetric FIR windows exhibit constant group delay with a linear phase profile across frequencies (Figure 31).

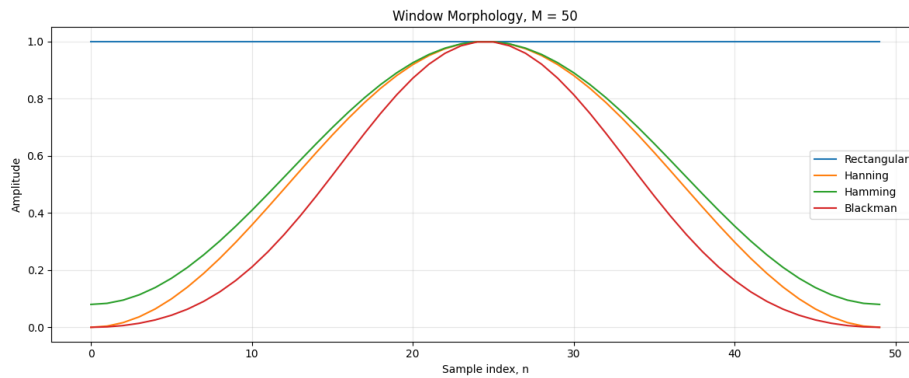


Figure 28: Window morphology for rectangular, Hanning, Hamming, and Blackman windows with $M = 50$.

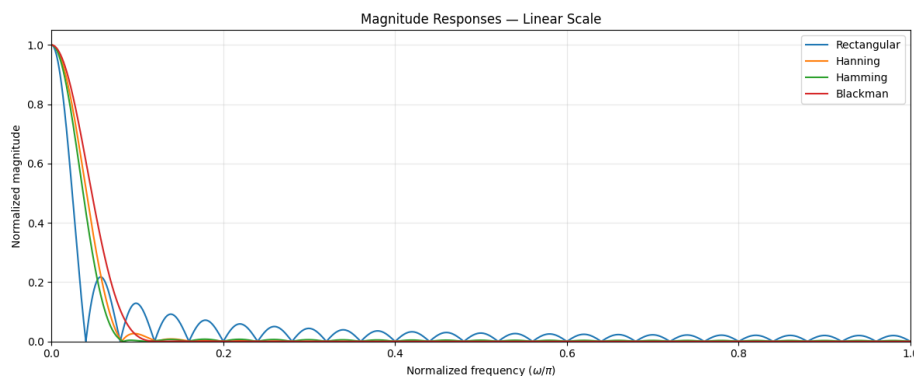


Figure 29: Magnitude response for rectangular, Hanning, Hamming, and Blackman windows with $M = 50$ (linear scale).

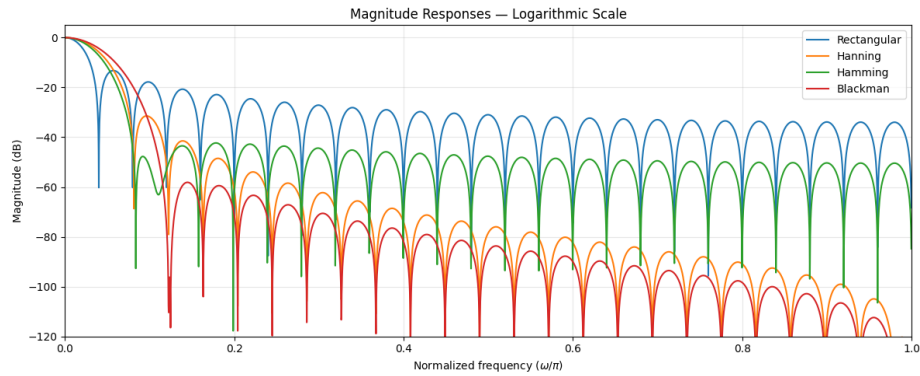


Figure 30: Magnitude response for rectangular, Hanning, Hamming, and Blackman windows with $M = 50$ (log scale).

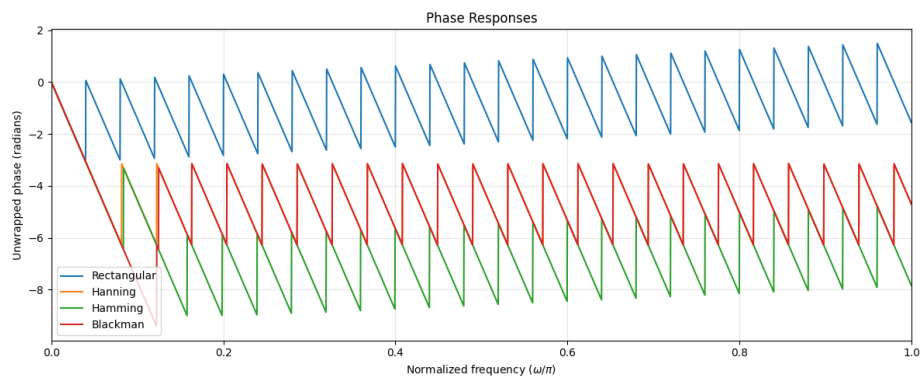


Figure 31: Phase response for rectangular, Hanning, Hamming, and Blackman windows with $M = 50$.

3.2 FIR Filter Design and Application Using the Kaiser Window

The Kaiser window is a flexible window function that approximates a variety of windows through a shaping parameter β and window length M . In this section, a Kaiser-windowed high-pass filter, a low-pass filter, and an FIR notch (comb) filter are designed to eliminate noise embedded within an ECG recording whose acquisition properties are outlined in Table 2.

Property	Value
Name of recording	ECG_with_noise.npz
Recorded Lead	Lead II
Sampling frequency	500 Hz
Amplitude units	mV

Table 2: Signal Properties for Noisy ECG Recording.

3.2.1 Time Domain and Power Spectral Density Analysis

The noisy ECG signal is first plotted in the time domain. Using `scipy.signal.periodogram`, the power spectral density (PSD) estimate is computed and analyzed to identify frequency content relative to normal ECG bandwidth (Figure 32). The main ECG signal content resides approximately between 0.05 Hz and 40 Hz, while overlapping high-frequency noise and power-line interference are visible.

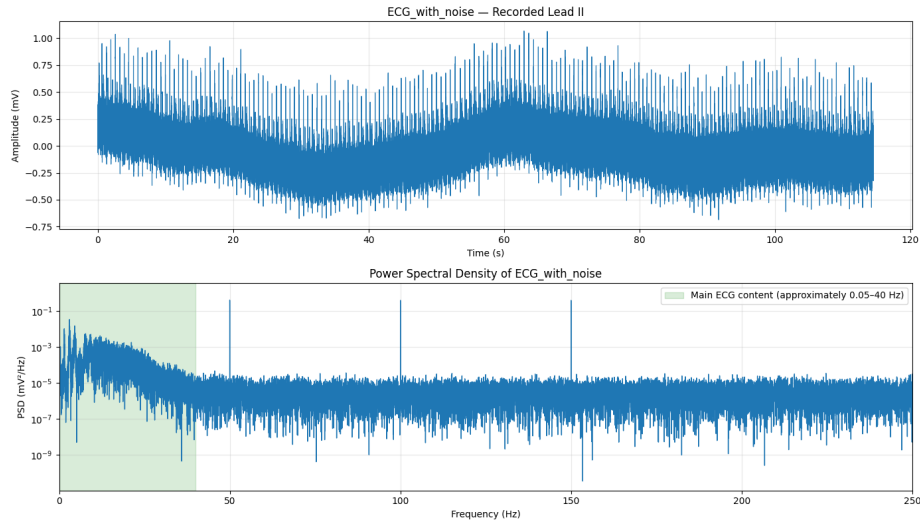


Figure 32: Original signal and Power Spectral Density of `ECG_with_noise` showing frequency content relative to normal ECG bandwidth.

3.2.2 Filter Specifications and Parameter Calculation

Based on the signal bandwidth and spectral inspection, filter parameters are defined with a stopband attenuation corresponding to a maximum amplitude error $\delta = 0.001$ ($A = 60$ dB):

1. **High-pass filter:** Passband begins at $f_{\text{pass}} = 5$ Hz, stopband ends at $f_{\text{stop}} = 1$ Hz ($\Delta f = 4$ Hz).
2. **Low-pass filter:** Passband ends at $f_{\text{pass}} = 125$ Hz, stopband begins at $f_{\text{stop}} = 140$ Hz ($\Delta f = 15$ Hz).
3. **Notch filter:** Target harmonic stopbands at $f_{\text{stop1}} = 50$ Hz, $f_{\text{stop2}} = 100$ Hz, and $f_{\text{stop3}} = 150$ Hz to eliminate power-line interference.

Using the empirical Kaiser window design formulas, a shaping parameter of $\beta = 5.6533$ is obtained alongside filter orders of $M = 453$ for the high-pass filter and $M = 121$ for the low-pass filter.

3.2.3 Filter Visualization

The Kaiser window profiles for the high-pass and low-pass filters are visualized in Figure 33. Furthermore, the magnitude and phase responses of the individual high-pass, low-pass, and notch FIR filters are displayed in Figures 34, 35, and 36, respectively.

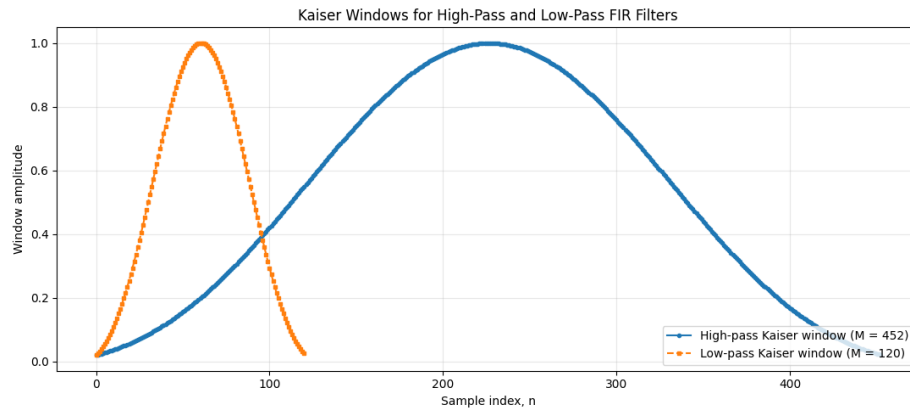


Figure 33: Kaiser windows for high-pass and low-pass FIR filters.

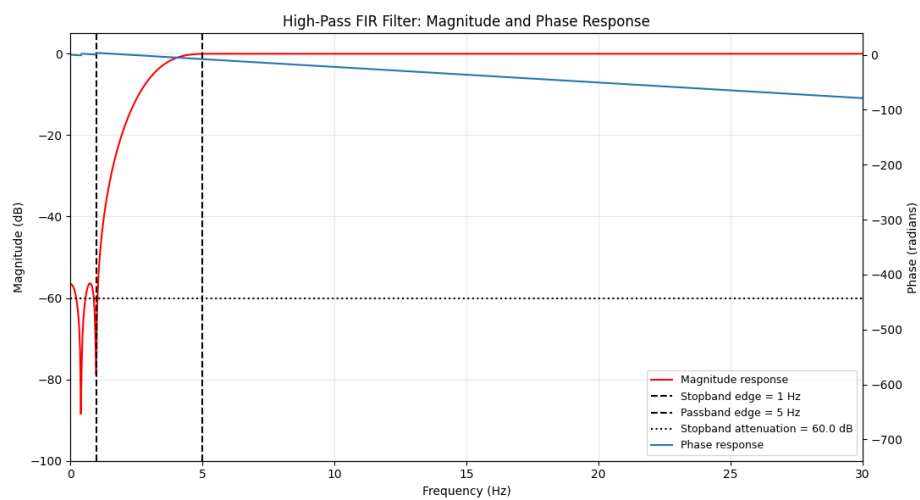


Figure 34: High-pass FIR filter: Magnitude and phase response.

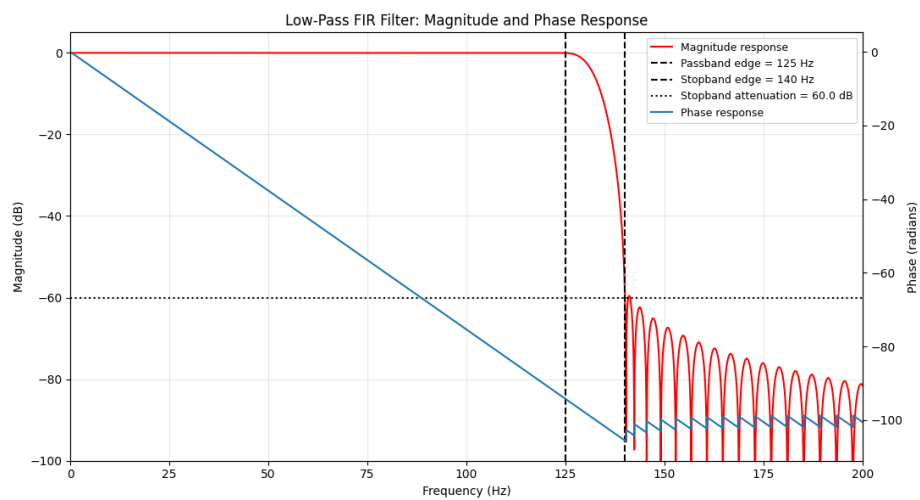


Figure 35: Low-pass FIR filter: Magnitude and phase response.

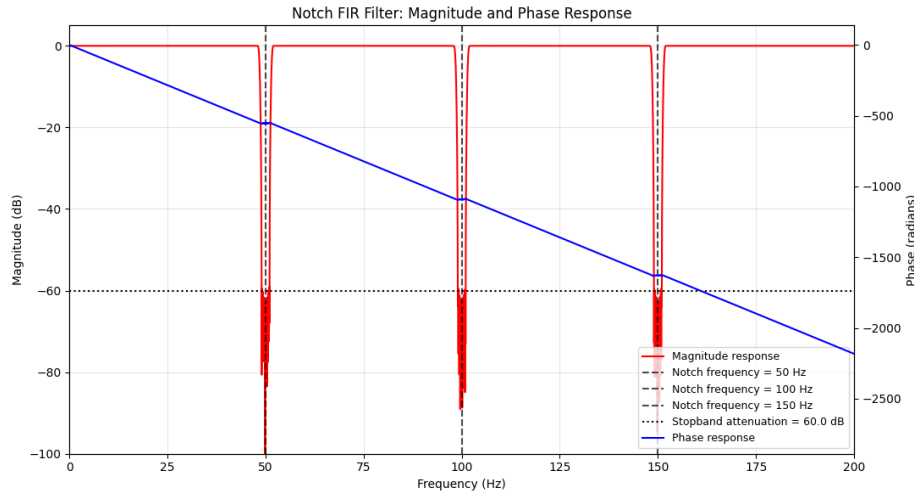


Figure 36: Notch FIR filter: Magnitude and phase response.

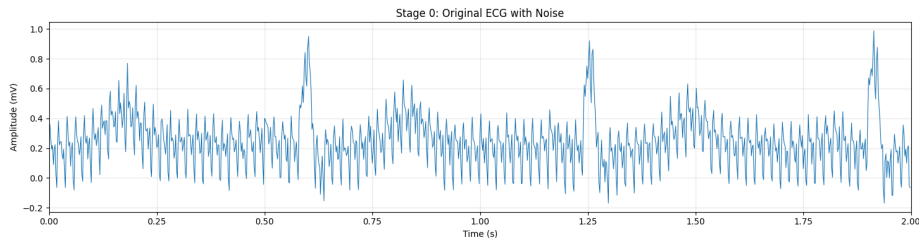


Figure 37: Stage 0: Original ECG signal corrupted with noise in the time domain.

3.2.4 Sequential Filter Application and Group Delay Compensation

The designed filters are applied sequentially to `ECG_with_noise.npz` with appropriate group delay compensation:

- **Stage 1 (High-pass):** Removes baseline wander, yielding the filtered output shown in Figure 38 with delay compensation of d_{hp} samples.
- **Stage 2 (Low-pass):** Cascades the low-pass filter to suppress high-frequency noise (Figure 39).
- **Stage 3 (Notch):** Applies the notch filter to eliminate 50 Hz, 100 Hz, and 150 Hz power-line interference harmonics (Figure 40).

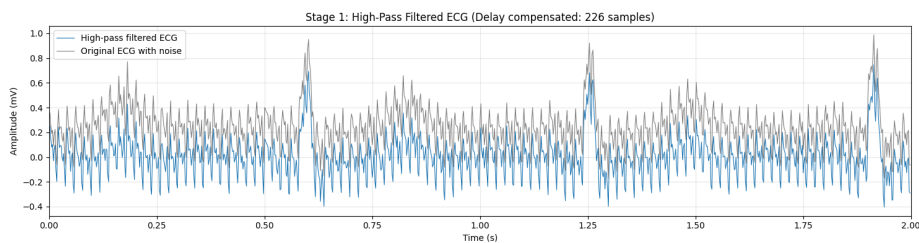


Figure 38: Stage 1: High-pass filtered ECG (Delay compensated).

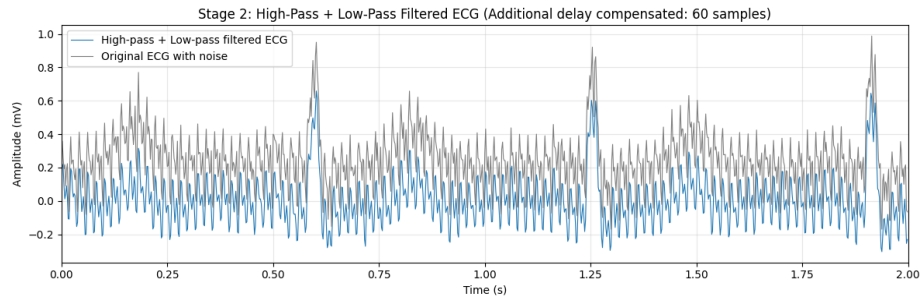


Figure 39: Stage 2: High-pass + Low-pass filtered ECG.

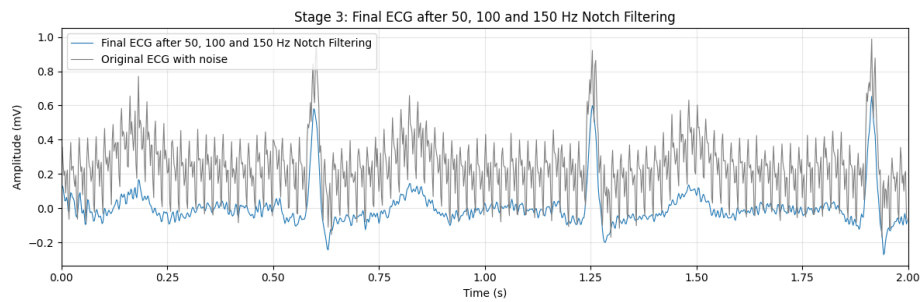


Figure 40: Stage 3: Final ECG after 50, 100, and 150 Hz notch filtering.

3.2.5 Combined Frequency Response and Final PSD

The magnitude response of the combined three-stage filter cascade is plotted in Figure 42, illustrating comprehensive passband preservation alongside deep attenuation at notch frequencies and stopbands. Finally, the PSD of the fully recovered ECG signal is presented in Figure 43, confirming successful noise suppression.

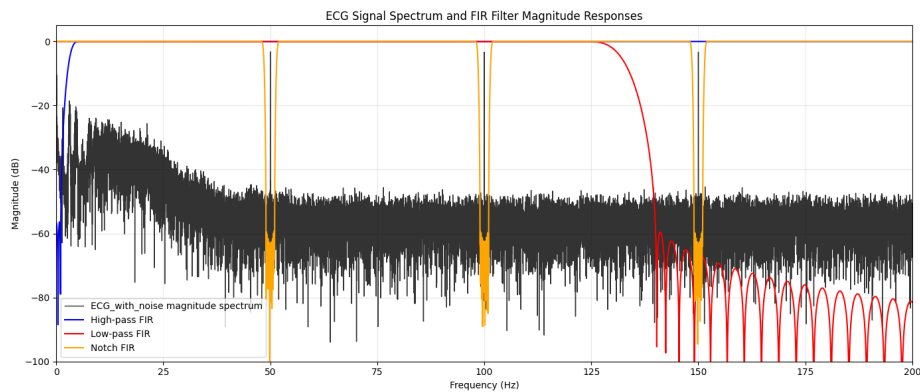


Figure 41: Magnitude responses of the high-pass, low-pass, and notch filters, together with the magnitude spectrum of the corrupted ECG signal.

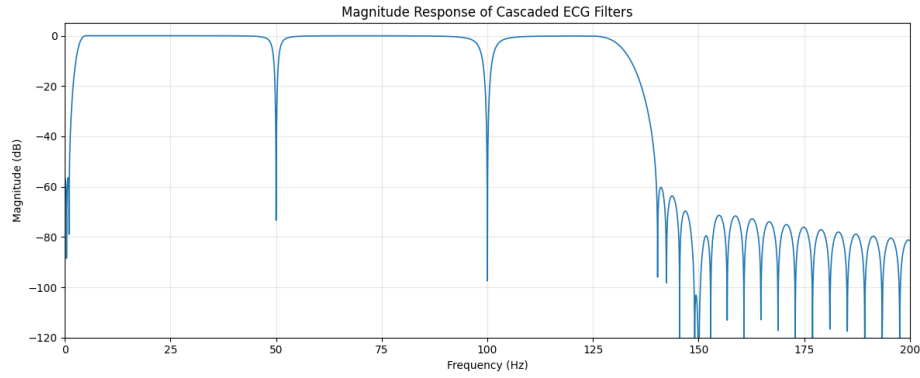


Figure 42: Magnitude response of cascaded ECG filters.

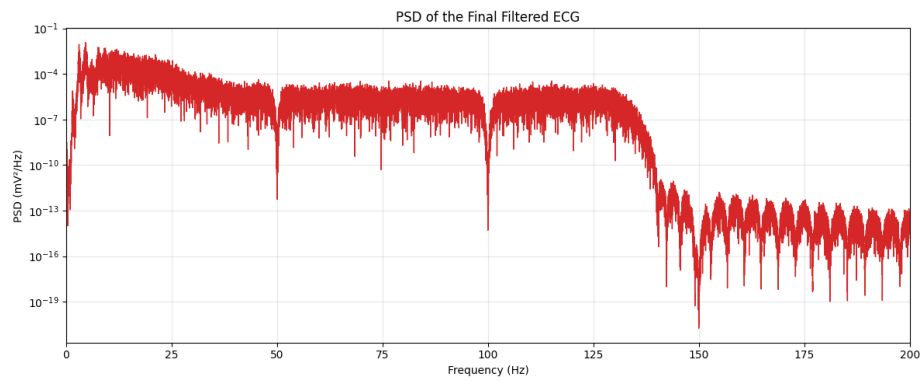


Figure 43: Power Spectral Density of the final filtered ECG signal.

4 IIR Filters

This section explores the realization of IIR Butterworth filters, the effects of their non-linear phase response, forward-backward filtering, and a comparison with the FIR filters implemented in the previous section.

4.1 Realizing IIR Filters

The realization of IIR Butterworth filters begins by leveraging the design parameters established in the previous section. For the low-pass filter, a Butterworth structure is created using an order of $M = 6$ and a cutoff frequency set to the midpoint of the transition band at 132.5 Hz. This yields the numerator coefficients $b = [0.0389, 0.2337, 0.5842, 0.7789, 0.5842, 0.2337, 0.0389]$ and denominator coefficients $a = [1.0000, 0.3561, 0.8250, 0.1712, 0.1266, 0.0116, 0.0020]$. The resulting magnitude response, phase response, and group delay of this low-pass IIR filter are illustrated in Figure 44

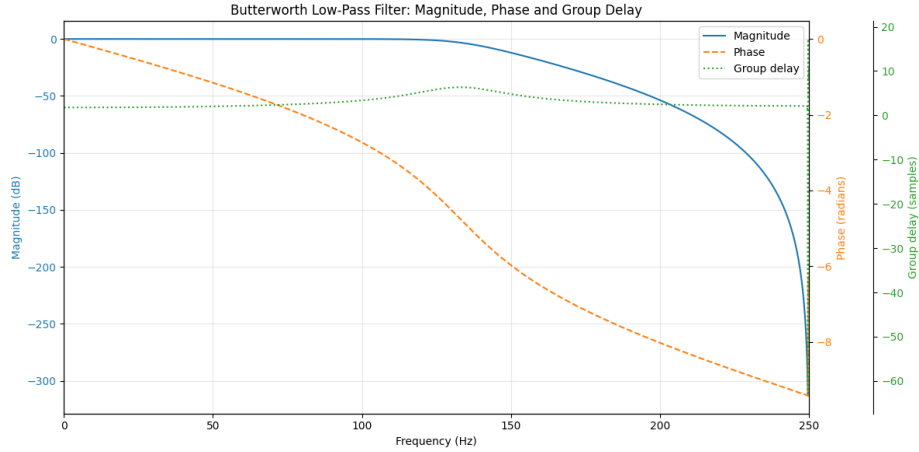


Figure 44: Magnitude, phase and group delay response of the low-pass IIR filter.

Following the low-pass design, a high-pass IIR filter is formulated using a passband edge of 5 Hz, a stopband edge of 1 Hz, and a maximum amplitude error $\delta = 0.001$, resulting in a filter order of 7 with an effective cutoff frequency of 3.21 Hz. The corresponding filter coefficients are determined as $b = [0.9134, -6.3936, 19.1809, -31.9682, 31.9682, -19.1809, 6.3936, -0.9134]$ and $a = [1.0000, -6.8188, 19.9292, -32.3629, 31.5359, -18.4401, 5.9910, -0.8343]$. Additionally, to target power-line interference, an IIR comb (notch) filter was constructed for a fundamental notch frequency of $f_0 = 50$ Hz and a quality factor of $Q = 30$, producing numerator coefficients $b = [0.9502, 0, \dots, -0.9502]$ and denominator coefficients $a = [1.0000, 0, \dots, -0.9004]$. The magnitude and phase responses for these high-pass and comb filter structures are captured in Figure 45 and Figure 46.

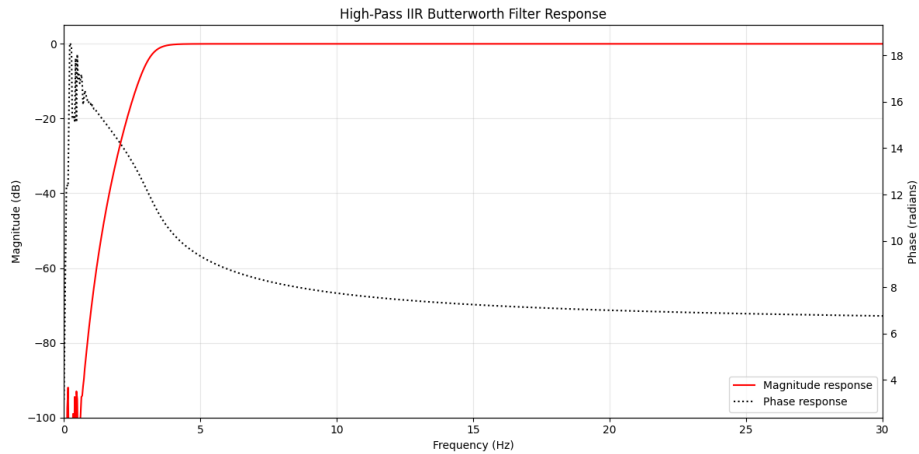


Figure 45: Magnitude and phase response of the high-pass IIR filter.

Additionally, to target power-line interference, an IIR comb (notch) filter was constructed for a fundamental notch frequency of $f_0 = 50$ Hz and a quality factor of $Q = 30$, producing numerator coefficients $b = [0.9502, 0, \dots, -0.9502]$ and denominator coefficients $a = [1.0000, 0, \dots, -0.9004]$. The magnitude and phase responses for these comb filter structures are captured in Figure 46.

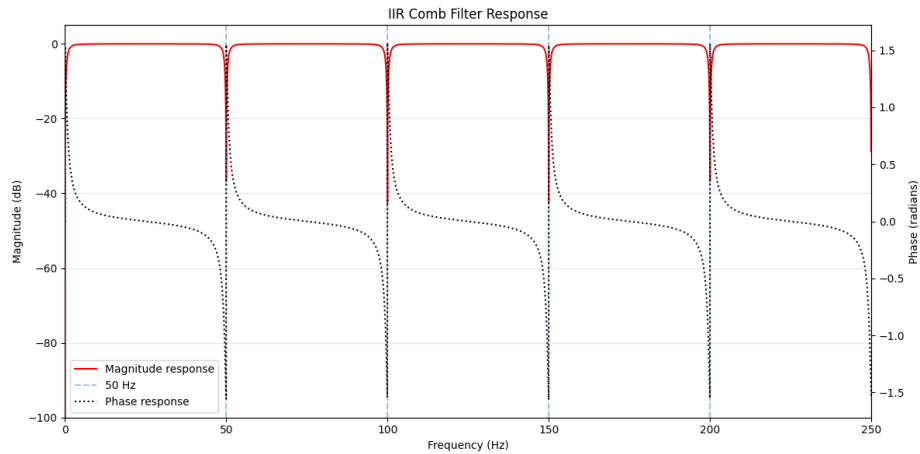


Figure 46: Magnitude and phase response of the IIR comb filter.

Finally, cascading the individual low-pass, high-pass, and comb IIR stages creates an overall filtering system whose combined magnitude response can be directly compared against the FIR filter cascade implemented in Section 3.2. As shown in Figure 47, the cascaded IIR architecture achieves sharp, efficient cutoffs and deep notch attenuations with significantly lower filter orders than its FIR counterpart, though at the expense of a non-linear phase profile.

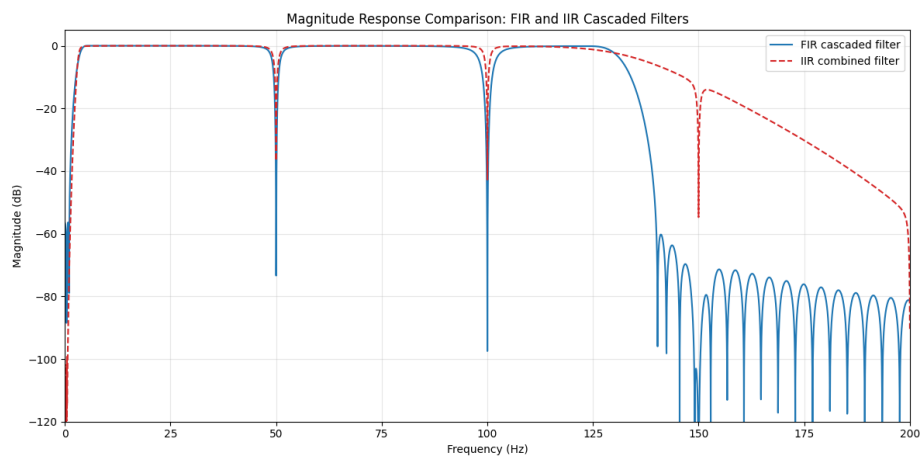


Figure 47: Magnitude response of the overall cascaded IIR filter compared to the FIR implementation.

4.2 Filtering Methods Using IIR Filters

When applying our cascaded IIR filter to `ECG_with_noise.npz` using standard forward filtering (`scipy.signal.lfilter()`), the resulting denoised signal exhibits noticeable phase shifts and morphological distortions (Figure 48). This behavior stems from the non-linear phase response and non-constant group delay characteristic of IIR filters.

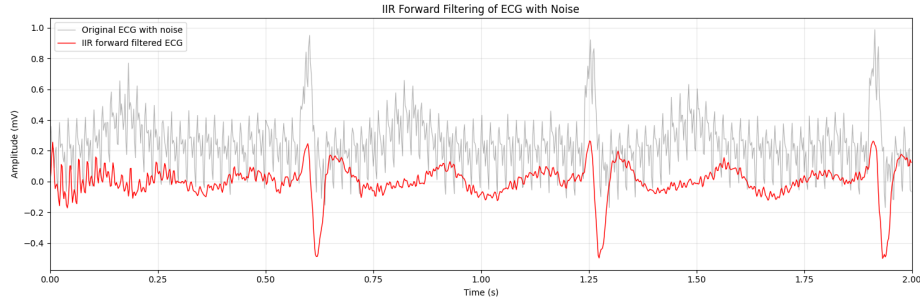


Figure 48: Denoised ECG signal after forward filtering with the overall IIR filter.

To counter this phase-induced distortion, we implement forward-backward filtering using `scipy.signal.filtfilt()`. By processing the signal in both forward and backward directions, the phase distortion is effectively cancelled out, yielding a zero-phase filtered output with preserved waveform features (Figure 49).

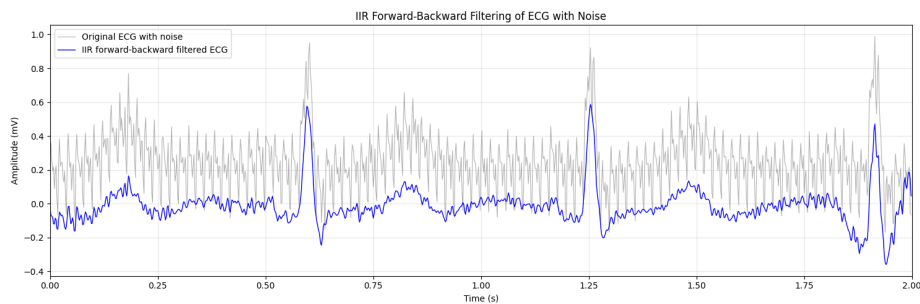


Figure 49: Denoised ECG signal after forward-backward filtering with the overall IIR filter.

For a direct performance comparison, Figure 50 zooms into a two-pulse window, overlapping the FIR-filtered signal from Section 3.2, the IIR forward-filtered signal, and the IIR forward-backward-filtered signal.

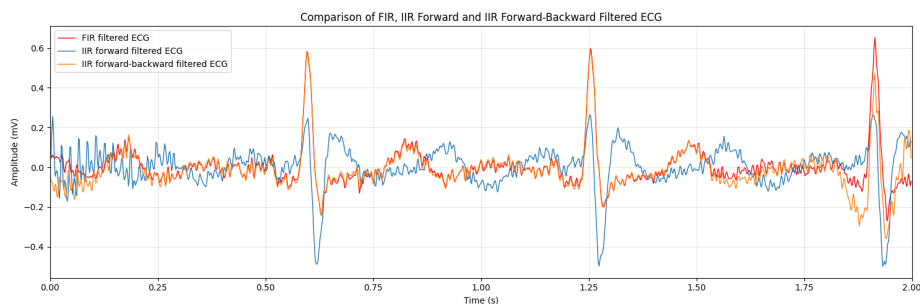


Figure 50: Comparison of FIR filtered signal, IIR forward-filtered signal, and IIR forward-backward-filtered signal over two ECG pulses.

The time-domain overlay highlights that while FIR filters deliver stable, distortion-free outputs thanks to linear phase (albeit requiring a high filter order), IIR filters achieve sharp attenuation at

lower orders. However, forward-only IIR filtering distorts the ECG morphology, whereas forward-backward IIR processing successfully eliminates phase errors to closely match the morphological fidelity of the FIR output.

Examining the spectral domain via the power spectral density (PSD) overlay of the three filtered signals (Figure 51) provides further insight:

- **FIR Filter:** Displays smooth, consistent attenuation with characteristic stopband ripples resulting from its finite window design.
- **IIR Forward Filter:** Exhibits spectral distortion and artifacts due to phase anomalies under single-pass implementation.
- **IIR Forward-Backward Filter:** Achieves a clean, flat stopband response combined with robust, deep attenuation across noise bands and comb notch frequencies.

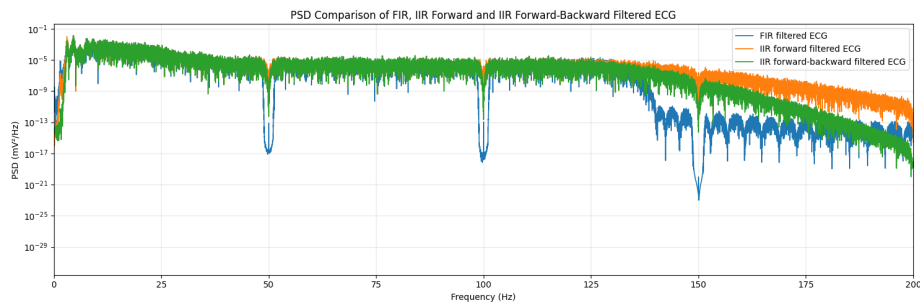


Figure 51: Power-spectral density plots of the FIR-filtered signal, IIR forward-filtered signal, and IIR forward-backward-filtered signal.

Ultimately, FIR filters guarantee stability and linear phase at the expense of computational weight and high order. IIR filters offer high efficiency and sharp cutoffs, though they risk non-linear phase distortion unless deployed via a forward-backward scheme, which effectively resolves phase errors by trading away real-time causality.